

Innovate, learn, sustain: unveiling dynamics of value addition in smallholder dairy farming

Muhammad Waqar Arshad and Muhammad Moazzam
*NUST Business School, National University of Sciences and Technology,
Islamabad, Pakistan*

Muhammad Mustafa Raziq
*College of Business Administration, University of Sharjah,
Sharjah, United Arab Emirates, and*

Waqas Ahmed
*NUST Business School, National University of Sciences and Technology,
Islamabad, Pakistan*

Journal of
Agribusiness in
Developing and
Emerging
Economies

83

Received 5 October 2023

Revised 11 March 2024

11 May 2024

Accepted 11 May 2024

Abstract

Purpose – This study explores value-added food products in smallholder dairy farming in developing countries by analyzing external pressures, supply chain learning, farmer innovation, education level, and food safety compliance.

Design/methodology/approach – We employed a quantitative approach by surveying 418 smallholder dairy farmers in three districts of Pakistan using interviewer-administered questionnaires. Data analysis involved confirmatory factor analysis and structural equation modeling.

Findings – The results indicate that external pressure significantly affects value-added smallholder dairy farms. This relationship is mediated by supply chain learning and farmers' innovative behavior, and moderated by farmers' education level and compliance with food safety standards.

Research limitations/implications – Further research is required to explore the drivers of value addition at the supply chain level.

Originality/value – This study contributes to the understanding of smallholder dairy farming dynamics and provides practical implications for improving value addition by managing the interplay between antecedents and promoting best practices in the industry.

Keywords Dairy farming, External pressure, Supply chain learning, Innovative behavior, Food safety standards, Value addition

Paper type Research paper

1. Introduction

The value proposition in food supply chains is multifaceted and linked to compliance with food safety standards from farm to fork (Wegerif and Martucci, 2019). These safety standards identify critical points in food production and supply chain processes and employ best practice measures to mitigate the risk of contamination or unsafe food production and handling for consumers (Gallo *et al.*, 2020). External pressure from customers, competitors, and regulators can be another factor that can affect the degree to which stakeholders adopt and implement such standards. Adetoyinbo and Mithöfer (2023) explored how resource-constrained small-scale farmers in developing countries utilize organizational networks to create and retain value in dynamic agri-food environments. Understanding the interplay between these external factors and their impact on businesses is crucial to enhancing the optimal implementation of food safety standards across the



Journal of Agribusiness in
Developing and Emerging
Economies

Vol. 16 No. 1, 2026

pp. 83-104

© Emerald Publishing Limited

2044-0839

DOI 10.1108/JADEE-10-2023-0250

Funding Statement: This research received no specific grant from any funding agency.

industry (Ahmed *et al.*, 2020a; Ahmed *et al.*, 2020b; Khan *et al.*, 2019). Owing to the rapid transformation of agri-food value chains driven by factors such as population growth, urbanization, increased consumer income, and heightened demand for food quality and safety, smallholder farmers in developing countries now require coordination to enhance farm performance owing to an increase in consumer demand for food safety (Abdul-Rahaman and Abdulai, 2020).

Organizing smallholder dairy farmers into groups enhances their position in the agri-food value chain, which is particularly crucial because of the perishability and limited shelf-life of food items with substantial demand across various regions (Negi and Trivedi, 2021). In today's competitive business environment, firms of all sizes strive to capitalize on market opportunities through proactive and innovative behaviors (Park *et al.*, 2019). Given the uncertain business landscape, companies must swiftly develop dynamic core competencies to effectively navigate ambiguity and uncertainty, enabling them to exploit market opportunities. Operational factors are the most significant contributors to supply chain inefficiency (Negi and Anand, 2019).

Political pressure significantly drives the acceptance of food safety standards, as indicated by studies (Ahmed *et al.*, 2018; Chen *et al.*, 2015; Fernando *et al.*, 2015; Khan *et al.*, 2019). New Zealand businesses embrace these standards to meet customer demand and enhance their market share (Chen *et al.*, 2015). Industry competition is another critical factor driving firms to adopt practices that distinguish their products (Escanciano and Santos-Vijande, 2014). In response to consumer pressure, firms adhere to food safety and quality standards to meet the rising consumer demand (Chen *et al.*, 2015; Escanciano and Santos-Vijande, 2014; Flores-Flores *et al.*, 2015). Consumer perception of food safety is multidimensional, influencing demand and prompting firms to implement these standards for added value (Jaeger *et al.*, 2023).

Smallholder dairy farming is crucial in many developing countries, contributing to food security, employment, and poverty reduction. Despite their significance, these farmers face challenges, such as low productivity, poor milk quality, and limited market access (Nyokabi *et al.*, 2021). Stakeholders, including governments, NGOs, and the private sector, intervene to address these issues, and often apply external pressure to change farmers' practices. Managing business relationships in agri-food chains with smallholders is more complex than in manufacturing industries (Owot *et al.*, 2023). One way to overcome these challenges is through value addition, involving processing and packaging to enhance milk quality, prolong shelf life, increase income, and boost competitiveness (Mottet *et al.*, 2020). However, engaging in value addition requires compliance with food safety and quality standards and demands knowledge and skills in these practices.

As defined by Haq (2021), supply chain learning involves acquiring and disseminating knowledge among stakeholders in a supply chain, including farmers, processors, traders, and retailers. For smallholder dairy farmers, embracing supply chain learning enhances proficiency in various areas, from production techniques to marketing and adherence to food safety regulations. Innovation is crucial in smallholder dairy farming as it requires farmers to adopt new practices to increase productivity, quality, and competitiveness (Nguyen-Anh, 2021). A cold chain is vital for building trust and profitability (Owot *et al.*, 2023). Lack of trust in local buyers increases transaction costs for farmers (Devaux *et al.*, 2018). Multi-stakeholder firms influence farmers' innovation processes with varying results (Barzola Iza *et al.*, 2020). Knowledge transfer within the supply chain significantly improves farmers' proficiency and performance (Haq *et al.*, 2021). Innovation, defined as adopting new practices, is critical for the success of smallholder dairy farming.

Farmers can showcase innovation by enhancing animal health, adopting new technologies, or creating new products. Innovation increases productivity, reduces costs, and improves market competitiveness. In today's competitive business environment,

startups and corporations actively pursue product market opportunities through proactive innovation (Park *et al.*, 2019). Given the uncertain and dynamic nature of the business landscape, firms must swiftly develop core emotional competencies to effectively navigate uncertainty and ambiguity and ensure that they can efficiently exploit market opportunities.

The literature underscores the pivotal role of innovative behavior in achieving business growth and developmental objectives. Small firms, however, lack well-explored mechanisms to stimulate innovation (Dunne *et al.*, 2016). External pressure, such as market demand and regulatory requirements, is suggested to drive farmers' innovative behavior and supply chain learning, leading to value addition in agriculture (Ahmed *et al.*, 2020c). The intricate relationship between external pressure, supply chain learning, innovative behavior, compliance with food safety standards, farmers' education, and their combined impact on value addition in smallholder dairy farming remains unclear. Understanding this dynamic relationship is crucial for developing effective business strategies for the agricultural sector.

Despite the growing focus on external pressure and its role in shaping smallholder farmers' behavior and learning processes, there is a gap in research regarding the collective impact of external forces, supply chain learning, farmers' innovative behavior, compliance with food safety standards, farmers' level of education, and value addition in smallholder dairy farming. Existing studies have primarily investigated individual components or singular interactions between external pressure and farmers' behavior, with little attention given to the combined effects of multiple factors on value addition in smallholder dairy farming. This study addresses this research gap by examining the intricate interactions between these variables and their combined effects on value addition in smallholder dairy farming in developing countries.

This study investigates the relationships between external pressure, supply chain learning, farmers' innovative behavior, and value addition in smallholder dairy farming. It explores how supply chain learning and creative behavior mediate the link between external pressure and value addition. The study also assesses how farmers' education level moderates the relationship between external pressure and innovative behavior and how compliance with food safety standards moderates the link between innovative behavior and value addition. These findings provide insights into the factors influencing value addition, policies, and interventions to enhance productivity and competitiveness in smallholder dairy farming. The article is structured into five sections, covering literature analysis; research gap identification; the proposed framework, methodology, and research results; and a discussion of the conclusions, contributions, limitations, and future research directions.

2. Literature review

This study investigates how external pressure affects value addition, with a farmer's innovative behavior and supply chain knowledge acting as mediators and a farmer's education and adherence to food safety regulations acting as moderators in the relationship between external pressure and value addition. Two well-established management theories, institutional isomorphism, sometimes known as institutional theory (DiMaggio and Powell, 1983), and resource-based views (Barney, 1991), are relevant to this study. According to the proponents of institutional theory, external forces may force a business to adopt a particular behavior (Ahmed *et al.*, 2020a; Ahmed *et al.*, 2020c; DiMaggio and Powell, 1983). Hence, institutional isomorphism is one of the most substantial explanations for farmer innovation, supply chain learning, and farmers' adoption of, or compliance with, minimal food safety standards.

2.1 External pressure and value addition

Numerous scholarly inquiries into smallholder dairy farming converge seamlessly with the foundational principles of the resource-based view, underscoring the paramount significance of tangible and intangible assets in shaping organizational performance. Government pressure, customer expectations, and industry competition, as outlined by DiMaggio and Powell (1983), significantly influence firms' production behavior. Government pressure compels businesses to adhere to ethical standards, particularly food safety, as mandated by laws and policies (Ab Talib *et al.*, 2016).

Various studies, including Korir *et al.* (2023), indicate that farm location, herd size, and education influence the adoption of dairy technology in smallholder farms. Snorek *et al.* (2023) underscored that diversification, knowledge sharing, and institutional support enhance resilience. According to NK and Uwimana (2023), contract farming positively affects smallholder dairy farmers' income. According to Penwell *et al.* (2023), crowd-collaborative farming is a way to optimize agricultural activities. In conclusion, external pressures positively impact value addition, aligning with the principles of the resource-based view. Therefore, we propose the following hypothesis:

H1. External pressure has a positive impact on value addition.

2.2 External pressure and Farmer's innovative behavior

Institutional theory posits that external pressure strongly influences organizational innovation. According to Doyle *et al.* (2014), firms facing increased regulatory pressure tend to adopt innovative practices to comply with the regulations. However, Lund-Thomsen *et al.* (2021) argued that pressure from global value chains could compromise compliance among cotton farmers in Pakistan and India. According to Mottet *et al.* (2020), government pressure is an "enabling environment," boosting smallholder farmers' innovative strategies in Cuba's food production system.

According to Najmi *et al.* (2023), they explored external pressure's impact on innovation in emerging markets, noting unique challenges like political instability and weak intellectual property rights. Firms that effectively navigate these pressures are more likely to engage in innovation, as observed by (Najmi *et al.*, 2023). According to Schiller *et al.* (2020), in Latin America, farmers' innovative behavior is crucial to the success of the agroecological transition. Overall, the literature suggests that external pressures, whether regulatory or institutional, significantly impact organizational innovation. Understanding and navigating these pressures could enable smallholder dairy firms to promote innovation and gain competitive edge. This hypothesis is formulated as:

H2. External pressure has a positive impact on farmers' innovative behavior.

2.3 External pressure and supply chain learning

External pressure is vital in shaping smallholder dairy farmers' learning behavior in supply chain management. Institutional theory and the theory of reasoned action reveal how regulatory factors, market demand, and environmental factors influence learning. As Doyle *et al.* (2014) demonstrated, regulatory pressure stimulates learning and innovation among firms, pushing smallholder dairy farmers to adopt best practices for safety and quality standards. Institutional pressure, market demand, and environmental factors affect learning behavior (Najmi *et al.*, 2023). The theory of reasoned action emphasizes that attitudes and subjective norms, influenced by external pressures such as peer networks, impact farmers' learning behavior. The literature collectively underscores that external pressures (Matzembacher and Meira (2019) significantly influence smallholder dairy farmers' supply chain learning behavior, and understanding these pressures can inform interventions to

enhance best practices and improve the sector's sustainability and competitiveness. This hypothesis is formulated as:

H3. External pressure has a positive impact on supply chain learning.

2.4 Farmer's innovative behavior and value addition

Various studies, including [Matzembacher and Meira \(2019\)](#) in Brazil; [Laguna et al., \(2021\)](#) in the Philippines; and [Erena et al. \(2022\)](#) in Ethiopia, highlight the significant impact of smallholder dairy farmers' innovative practices on value addition. In Brazil, innovative production processes have led to higher value-added activities and product prices. Similarly, in the Philippines, improved animal genetics and feeding practices correlated with higher milk yields, enabling greater engagement in value addition. [Lewison \(2022\)](#) emphasized the importance of farmers' entrepreneurial strategies and value addition for inclusive value chains in Nepal. A recent study in Pakistan by [Najmi et al. \(2023\)](#) further confirmed a positive association between smallholder dairy farmers' innovative behavior and participation in value-adding activities, correlating with higher incomes and increased resilience to market shocks. The literature collectively emphasizes the pivotal role of farmers' innovative behavior in driving value addition, enhancing productivity, improving milk quality, and reducing costs. Policies and interventions should prioritize fostering farmers' innovation and supporting collective action in smallholder dairy farming to promote value addition. Therefore, the following hypothesis can be formulated:

H4. Farmer's innovative behavior has a positive impact on value addition.

2.5 Supply chain learning and value addition

Academics and industry experts recognize supply chain learning for its potential to enhance supply chain management (SCM) systems ([Huo et al., 2019](#); [Yang et al., 2019](#)). This process involves acquiring, integrating, and applying knowledge from internal operations, suppliers, and clients, fostering adaptability and enhancing value addition ([Huo et al., 2019](#); [Yang et al., 2019](#)). Firms emphasizing the importance of SCM learning for profitability prioritize information sharing with partners ([Shen et al., 2019](#)). In the context of smallholder dairy farmers, supply chain learning can improve compliance with safety standards, influence attitudes towards value addition, and provide valuable knowledge and resources. Policymakers and practitioners should prioritize interventions tailored to farmers' needs, including training in milk quality, safety, animal health, nutrition, market access, and new technologies related to value addition. Therefore, we can hypothesize.

H5. Supply chain learning has a positive impact on value addition.

2.6 Farmer's innovative behavior as a mediator

Although the literature lacks extensive exploration, the mediating role of smallholder farmers' innovative behavior amid external pressure and value addition has gained traction in recent studies. According to [Yin et al. \(2022\)](#), farmers' innovative behavior mediates the relationship between external pressures (such as market competition) and value addition. [Agyabeng-Mensah et al. \(2023\)](#) find a similar mediation of institutional pressure in the Ghanaian dairy industry. Previous research supports that a firm's resources, including farmers' innovative behavior, contribute to a sustained competitive advantage ([Haq, 2021](#)). Institutional theory asserts that external pressures such as regulations and norms influence actors' innovative behavior within an industry, including smallholder farmers ([Haq et al. \(2021\)](#)). This emphasizes the need to promote and support farmers' innovative behavior to

enhance their competitiveness and sustainability in the market. Hence, we propose the following hypothesis:

- H6. Farmer's innovative behavior mediates the relationship between external pressure and value addition.

2.7 Supply chain learning as a mediator

Supply chain learning is pivotal in today's competitive business landscape as it enhances farmers' skills and knowledge, increasing their capacity to create value. External pressures, including the government, customers, and normative pressures, significantly affect supply chain learning and value addition. However, these external pressures mediate the relationship between value addition and supply chain learning.

Applying the resource-based view (RBV), [Ojha et al. \(2018\)](#) found that supplier innovation positively influences supply chain learning and enhances value addition. Similarly, using institutional theory, [Ahmed et al. \(2020b\)](#) discovered that environmental pressure positively influenced supply chain learning, which enhanced value addition. According to [Haq \(2021\)](#), employing the RBV establishes that customer pressure positively influences supply chain learning, contributing to increased value addition. In summary, supply chain learning is a crucial mediator between external forces (the government, customers, and normative forces) and value addition. [Zhang and Lv \(2015\)](#) emphasized that investing in skills and knowledge is essential for farmers to enhance value addition and increase farm yield, profit, and income. This hypothesis is formulated as:

- H7. Supply-chain learning mediates the relationship between external pressure and value addition.

2.8 Farmer's level of education as a moderator

The variables influencing the innovative behavior of smallholder dairy farmers have been the subject of numerous studies. According to [Liu et al. \(2021\)](#) and [Tan et al. \(2021\)](#), education plays a crucial role in encouraging entrepreneurial behavior and lessening the effects of external shocks. Socioeconomic characteristics and entrepreneurial behavior shape farmers' market orientation and marketing arrangements ([Mandala et al., 2021](#); [Wangu et al., 2021](#)). Similarly, the interrelationship between trust, market information, and innovative behavior has been investigated by various researchers ([Mugonya et al., 2021](#); [Nurlaela et al., 2022](#); [Xhoxhi et al., 2019](#)). [Mellon Bedi et al. \(2022\)](#) highlighted the potential of information technology and interventions to promote and assist smallholder dairy farmers' creative practices. The adoption of innovations in dairy farming by smallholder dairy farmers is highly influenced by their educational levels ([Chaurasiya et al., 2017](#); [Manjera, 2021](#)). Nevertheless, this study does not explicitly address how education level modifies the link between external pressure and innovation behavior. Further research is needed to investigate this moderating effect. Therefore, the following hypothesis is proposed:

- H8. The farmer's education moderates the relationship between external pressure and the farmer's innovative behavior.

2.9 Compliance with food safety standards as a moderator

The relationship between farmers' innovative behavior and value addition can be moderated by compliance with food safety requirements. Research conducted by [Walder et al. \(2019\)](#) and [Jatib \(2003\)](#) emphasized the significance of innovation in agriculture, with the latter underlining the role of food safety and quality assurance in boosting competitiveness. Individual perceptions and preferences, the impact of institutional structures, and risk-

related latent factors affect the adoption of these technologies (Kurihara *et al.*, 2014; Roussy *et al.*, 2017). Takács-György (2016) highlights the significance of innovation in food safety and the application of managerial abilities and teamwork in creating solutions. Consequently, compliance with food safety regulations is only one of several elements that impacts the intricate relationship between a farmer's creative practices and value addition, even though it can moderate it. Further research is required to examine this potential moderating effect. Therefore, the hypothesis is given as:

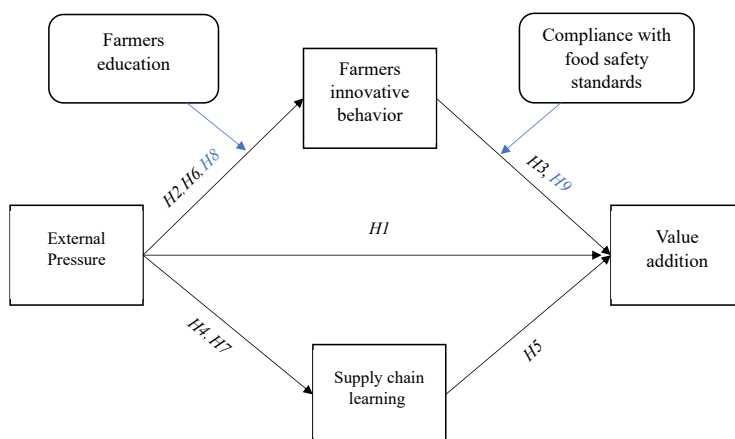
H9. Compliance with food safety standards moderates the relationship between farmers' innovative behavior and value addition.

Contemporary literature on the nexus between external pressure, farmers' innovative behavior, supply chain learning (SCL), and farmers' value addition in the dairy sector of an emerging economy is scarce. Although a few studies by Najmi *et al.* (2023) have examined external pressure, including government regulations, market demands, and customer preferences, the literature fails to investigate the possible mediating effect of farmers' innovative behavior and supply chain learning as well as the moderating impact of farmers' education levels and compliance with food safety standards.

This study aims to bridge this research gap by investigating the influence of external pressure (EP) on value addition (VA) in Pakistan's dairy sector, focusing on the mediating role of farmers' innovative behavior and supply chain learning (SCL). Furthermore, this research seeks to contribute to the literature by exploring how farmers acquire and utilize knowledge from various sources such as suppliers and customers to enhance the value of their products. Additionally, this study examines the potential moderating impact of farmers' education level, whether farmers with education have a more significant effect on innovative behavior than those without education, and compliance with food safety standards (FSS) on the relationship between external pressure (EP) and value addition (VA). The proposed framework is illustrated in Figure 1.

3. Methodology

We adopted a quantitative methodology to collect cross-sectional data from smallholder dairy farmers. Pakistan is the fourth largest milk-producing country (64.33 million tonnes) in



Source(s): Authors' own work

Figure 1.
Proposed analytical
framework

the world, and its national dairy herd is predominantly a smallholder enterprise (FAO, 2024). Data were collected by visiting smallholder dairy farmers in four districts of Pakistan, Khanewal, Lodhran, Muzaffargarh, and Layyah, located around the Indus River Delta. Due to the non-availability of the sampling frame, a purposive sampling technique was used to select a sample of 418 smallholder farmers with fewer than eight dairy animals, who were interviewed face-to-face by the research team to fill out the survey questionnaires. After treating the data for missing values (Kock, 2015), 418 responses were included in the final analysis. The sample size used in our study exceeds the minimum requirement calculated using G*Power software (Ramayah *et al.*, 2018). Moreover, various tests of the reliability and validity of the data were performed to mitigate common method bias. The questionnaire used in this study comprised close-ended questions covering all research constructs. A 10-item scale modified by Shea *et al.* (2014) was adopted to comply with food safety standards. The 15-item scale developed by Haq *et al.* (2021) was used. A 23-item scale was adapted for supply chain learning to measure value addition (Tavana *et al.*, 2019). An 11-item scale has been adopted, modified, and used to measure external pressure from Cao and Chen (2017) and Huang *et al.* (2013).

The demographic variable farmers' educational level was used to moderate the relationship between external pressure and value addition. The responses of the farmers were measured using a five-point Likert scale, where "Strongly Disagree" was given a score of 1, "Disagree" a score of 2, "Neither Agree nor Disagree" a score of 3, "Agree" a score of 4, and "Strongly Agree" a score of 5. To test moderation, farmers' education was divided into two groups: one with no education and the other with education, and a multi-group analysis was performed (Memon *et al.*, 2019).

Structural equation modeling (SEM), a second generation statistical technique, was employed to investigate the complex structural relationships between the selected variables (Hair Jr *et al.*, 2021). According to Hair Jr *et al.* (2021), SEM is used to overcome the three limitations of 1st generation statistical techniques, such as multiple regression, logistic regression, and analysis of variance. Their limitations are (1) the postulation of a simple model structure involving one layer of dependent and independent variables, i.e. causal chains, such as "A leads to B leads to C" and so on, (2) requiring that all variables can be considered observable, and (3) the assumption that all variables are measured without error. Structural equation Modeling (SEM) is either used to test (accept/reject hypotheses) theories called Covariance-based SEM or to establish "causal-predictive" associations between constructs or variables called partial least squares (PLS-) SEM. This study uses PLS-SEM to simultaneously model and estimate relationships among multiple dependent and independent variables, comprising the measurement and structural models (Becker *et al.*, 2023).

Moreover, PLS-SEM is particularly well-suited to investigating complex relationships and interactions among variables and is best suited to the research context of this study (Hair Jr *et al.*, 2021; Sarstedt *et al.*, 2023). A prerequisite of PLS-SEM is confirmatory factor analysis, also called the measurement model, which tests the correlations between outer loadings (variables) and inner loadings (constructs) (Hair Jr *et al.*, 2021). Lastly, ex-ante measures, such as pilot testing of the questionnaire, project brief, informed consent, anonymity, and confidentiality, and post-ante measures, such as testing normality, validity, and reliability of the data collected through the survey, were performed to overcome the common method bias and respondent bias.

4. Results and discussion

4.1 Demographics

Table 1 presents the demographic characteristics of the 418 dairy farmers, revealing that many lacked formal education, with 61% having no educational background, 29.50%

Table 1.
Dairy farmer's
education, experience,
and income level

	N	Percentage	Cumulative percentage
<i>Education</i>			
No formal education	255	61	61
1–10 years	123	29.50	90.50
11–12 years	16	3.80	94.30
13–16 years	14	3.40	97.70
Above 16 years	10	2.30	100
<i>Experience</i>			
Less than five years	44	10.60	10.60
6–10 years	131	31.40	42
11–15 years	133	31.80	73.80
16–20 years	68	16.20	90
Above 20 years	42	10	100
<i>Income</i>			
PKR 20,000 – PKR 40,000	17	4.2	4.2
PKR 40,001 – PKR 60,000	89	21.2	25.4
PKR 60,001 – PKR 80,000	193	46.2	71.6
PKR 80,001 – PKR 100,000	98	23.5	95.1
Above PKR 100,000	21	4.9	100
Source(s): Authors' own work			

receiving formal education ranging from one to ten years, and the remaining respondents with academic qualifications of 11–16 years or more. The data suggest that many smallholder farmers in Pakistan have limited access to formal education, resulting in low literacy rates in rural areas, and insufficient knowledge and understanding of supply chain practices.

Most dairy farmers surveyed earned an average income of 60,000–80,000, with some earning less because of limited knowledge and resources. This study highlights the influence of external pressures such as market demands, regulations, and consumer preferences on smallholder dairy farmers' decisions, emphasizing their impact on learning, innovation, and compliance with food safety standards, ultimately contributing to value addition in the industry. Pakistan's dairy production, predominantly led by smallholders, mirrors many underdeveloped subcontinental nations. The prevalence of small-scale, subsistence, or market-oriented farming, with a significant market share held by peri-urban or commercial farmers, aligns with the broader trend in developing nations (Sarwar *et al.*, 2002). Except for peri-urban areas, most dairy operations are fragmented and subsistence-based, emphasizing the importance of understanding the dynamics of the group supplying 97% raw/fresh milk. Focusing on this group is vital as raw/fresh milk constitutes a substantial part of total consumption in the tested areas (Tahir *et al.*, 2019). By understanding the actions of these major milk suppliers, policymakers can gain insights crucial for developing policies that enhance sustainability, efficiency, and safety in Pakistan's dairy sector. Recognizing the current conditions in the dairy farming sector is essential given the private sector's predominant control in urban and rural settings.

Overall, the findings highlight the need for targeted interventions to improve smallholder farmers' education and knowledge of modern dairy practices and supply chain management to enhance their livelihoods and boost the growth of Pakistan's dairy sector.

4.2 Measurement model

A two-step procedure was employed following the proposed PLS-SEM path. The first stage of this procedure involves assessing the measurement model and the second stage involves

assessing the structural model. *Post-ante* measures of common method bias include testing the normality, validity, and reliability of data collected through the survey. According to the results presented in Table 2, the composite reliability scores for external pressure (0.938), farmers' innovative behavior (0.917), supply chain learning (0.932), and value addition (0.854) demonstrated high internal consistency, as they fell within the acceptable range according to the criteria established by Hair Jr et al. (2021).

The adequacy of discriminant validity was established by comparing the square root of the average variance extracted (AVE) to the correlation between the latent variables, as proposed by Fornell and Larcker (1981). Moreover, all factor loadings were subjected to confirmatory factor analysis, and any factor loadings below the threshold point of 0.50 were removed. Consequently, factor loadings ranging from 0.794 to 0.911 (Table 3) were retained based on the guidelines provided by Hair Jr et al. (2021). Hence, it can be established that the data used for further analysis was free from the common method bias.

Both the linear relationship and mediation analysis models were subjected to thorough scrutiny of their structural components. Five potential direct associations were tested, and the results are listed in Table 4. Of the five tested hypotheses, only one failed to meet the necessary significance level and *t*-value. Analyses were conducted via bootstrapping, using 418 cases/observations and 10,000 iterations.

Table 5 presents the findings on farmers' innovative behavior and supply chain learning as mediators in the relationship between external pressure and value addition. This study establishes that the association between external pressure and value addition is entirely mediated by farmers' innovative behavior, as evidenced by the significant β coefficient of 0.181, $t = 5.113$ ($p < 0.001$).

Furthermore, supply chain learning exhibited complete mediation between external pressure and value addition, with a significant β coefficient of 0.127 and $t = 2.047$ ($p < 0.001$). These results signify the crucial role of farmers' innovative behavior and supply chain learning in facilitating value addition when facing external pressures.

The measurement model presented in Figure 2 provides evidence of the mediating influence of farmers' innovative behavior and supply chain learning on the relationship between external pressure and value addition. The measurement properties of the model were evaluated using factor loadings as the indicators. The findings revealed that factor loadings below the recommended threshold of 0.50 were excluded from the analysis, as stipulated by established guidelines, were excluded from the analysis. As all the variables are reflective, the items with average variance extracts below 0.5 were deleted to achieve convergent and discriminant validity (Hair Jr et al., 2021). Conversely, the factor loadings that surpassed this threshold, as detailed in Table 3, ranged from 0.714 to 0.911 and were considered acceptable for further analysis. The results demonstrate the valuable contributions of farmers' innovative behavior and supply chain learning as mediators that connect external pressure and value addition.

Table 2.
Mean, SD, composite
reliability, correlations

	<i>M</i>	<i>SD</i>	<i>CR</i>	1	2	3	4
1. External pressure	4.08	0.602	0.938	0.832			
2. Farmers innovative behavior	4.40	0.081	0.917	***	0.887		
3. Supply chain learning	4.21	0.797	0.932	**	**	0.813	
4. Value addition	3.90	0.613	0.901	*	*	*	0.833

Note(s): CR = composite reliability, * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$

Source(s): Authors' own work

Constructs	Items	Loadings	Cronbach's alpha	Composite reliability	AVE
External pressure	EPR1	***	0.888	0.938	0.692
	EPR2	***			
	EPR3	***			
	EPR4	***			
	EPR5	***			
	EPR6	0.787			
	EPR7	***			
	EPR8	0.842			
	EPR9	0.836			
	EPR10	0.851			
	EPR11	0.842			
Farmers innovative behavior	FIN1	0.911	0.864	0.917	0.786
	FIN2	0.900			
	FIN3	0.848			
	FIN4	***			
	FIN5	***			
	FIN6	***			
	FIN7	0.753			
Supply chain learning	SCL1	0.814	0.915	0.932	0.662
	SCL2	***			
	SCL3	***			
	SCL4	***			
	SCL5	***			
	SCL6	0.800			
	SCL7	0.820			
	SCL8	***			
	SCL9	***			
	SCL10	0.794			
	SCL11	***			
	SCL12	0.842			
	SCL13	0.826			
	SCL14	0.797			
	SCL15	***			
Value addition	Vad1	***	0.853	0.901	0.695
	Vad2	***			
	Vad3	***			
	Vad4	***			
	Vad5	***			
	Vad6	***			
	Vad7	***			
	Vad8	***			
	Vad9	***			
	Vad10	***			
	Vad11	***			
	Vad12	***			
	Vad13	0.830			
	Vad14	0.820			
	Vad15	***			
	Vad16	***			
	Vad17	0.821			
	Vad18	0.861			
	Vad19	***			
	Vad20	***			
	Vad21	***			
	Vad22	***			
	Vad23	***			

(continued)

Table 3.
Confirmatory factor
analysis and factor
loadings

Table 3.

Constructs	Items	Loadings	Cronbach's alpha	Composite reliability	AVE
Compliance with food safety standards	CWFSS1	0.887	0.921	0.938	0.717
	CWFSS2	0.863			
	CWFSS3	0.814			
	CWFSS4	0.835			
	CWFSS5	0.846			
	CWFSS6	***			
	CWFSS7	***			
	CWFSS8	***			
	CWFSS9	***			
	CWFSS10	0.835			

Note(s): α = Cronbach's alpha; *** item deleted during factor analysis
Source(s): Authors' own work

Table 4.

Results of hypothesis testing (direct effects)

Direct relationships	<i>B</i>	SD	<i>T</i> stats	<i>p</i> -value	Decision	<i>R</i> ²
EP → VA	0.008	0.067	0.123	0.902	Not confirmed	
EP → FIN behavior	0.504	0.050	10.08	0.000	Confirmed	0.486
FIN behavior → VA	0.359	0.063	5.707	0.000	Confirmed	0.680
EP → SCL	0.830	0.020	41.41	0.000	Confirmed	0.689
SCL → VA	0.160	0.053	3.03	0.000	Confirmed	0.680
Far Lev Edu × EP → FIN behavior	0.121	0.052	2.312	0.000	Confirmed	
CWFSS × FIN behavior → VA	−0.103	0.032	3.187	0.000	Confirmed	

Note(s): *B* = beta value, EP = external pressure, FIN behavior = farmers' innovative behavior, SCL = Supply chain learning, VA = value addition, CWFSS = compliance with food safety standards *** $p < 0.001$
Source(s): Authors' own work

Table 5.

Test of mediation

Indirect paths	<i>B</i>	SD	<i>T</i> stats	<i>p</i> -value	Findings
External pressure → farmers innovative behavior → value addition	0.181	0.035	5.113	0.000	Full mediation
External pressure → supply chain learning → value addition	0.127	0.062	2.047	0.041	Full mediation

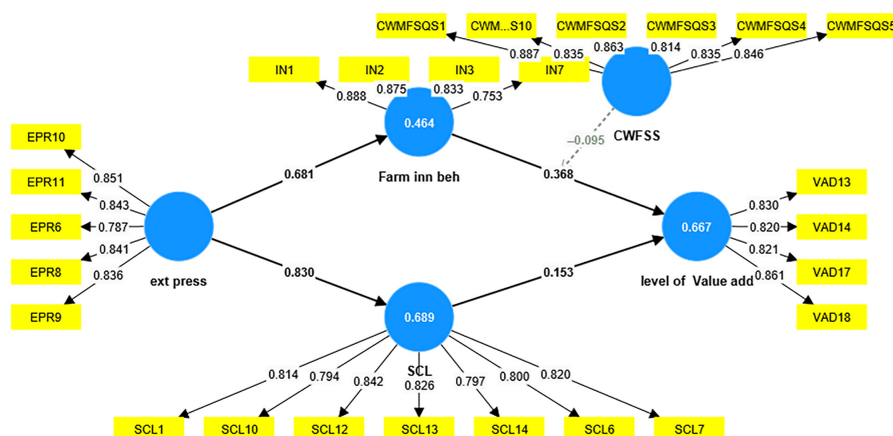
Note(s): *B* = beta value
Source(s): Authors' own work

Table 4 presents the findings of farmers' level of education moderating the relationship between external pressure and innovative farmer behavior ($\beta = 0.121, t = 2.312, (p < 0.001)$).

Table 6 shows that compliance with food safety standards moderates the relationship between farmers' innovative behavior and value addition ($\beta = -0.103, t = 3.187 (p < 0.001)$). The structural model depicted in Figure 3 includes moderating variables, namely the farmers' education level and adherence to food safety standards, both of which are supported.

4.3 Discussion

This study addresses the literature gap by empirically examining the interactions between external pressure, farmers' innovative behavior, supply chain learning, and value addition. It contributes to supply chain learning research, adds to the body of knowledge, and fulfills the



Source(s): Authors' own work

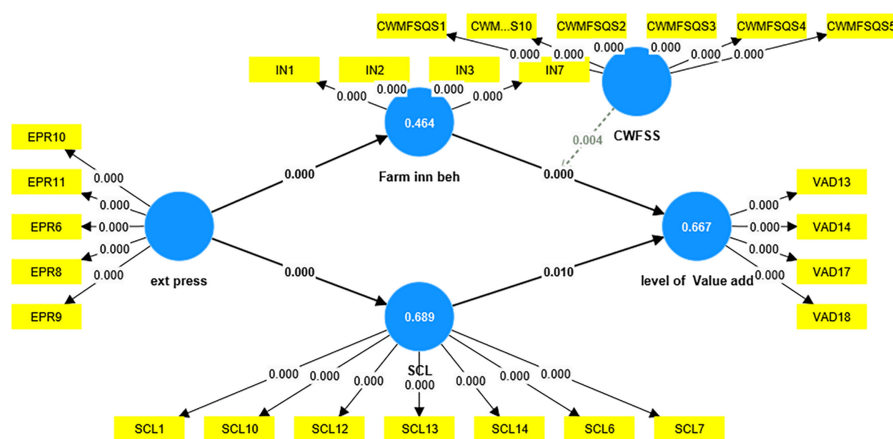
Figure 2.
Confirmatory factor
analysis and factor
loadings

Indirect paths	<i>B</i>	SD	<i>T</i> stats	<i>p</i> -value	
1. CWFSS × FIN behavior → VA	−0.103	0.032	3.187	0.041	Supported

Note(s): *B* = beta value

Source(s): Authors' own work

Table 6.
Test of moderation



Source(s): Authors' own work

Figure 3.
Structural model with
moderating variables

need for empirical studies on aiding factors in SC learning (Thöni and Tjoa, 2017). The first hypothesis, regarding the direct impact of external pressure, was insignificant ($\beta = 0.008$, $t = 0.123$, $p < 0.001$). However, this study reveals how external pressures, including consumer demands and regulations, drive smallholder businesses to adopt proactive food safety practices and add value. Government intervention is crucial to ensure food safety and

promote value addition in Pakistan. The impact of external pressure on farmers' innovative behavior was significant ($\beta = 0.504, t = 10.08, p < 0.001$), aligning with the interdependence of internal and external forces on innovation. Hypothesis 3 confirmed a positive relationship between farmers' innovative behavior and value addition ($\beta = 0.359, t = 5.707, p < 0.001$).

This study aligns with previous research by Tebeka *et al.* (2017) and Mukhtar *et al.* (2018), highlighting the essential role of innovative behavior in value addition to farm products and services. Innovative behavior involves risk-taking, decision-making, and information gathering, which are crucial for meeting customer needs. Wongnaa *et al.* (2019) emphasized its significance in promoting economic prosperity, poverty reduction, and improving agricultural quality of life. Hypothesis 4 confirmed a positive relationship between external pressure and supply chain learning ($\beta = 0.830, t = 41.41, p < 0.001$). Hypothesis 5 indicates that supply chain learning positively impacts value addition ($\beta = 0.160, t = 3.03, p < 0.001$). Hypotheses 6 and 7 suggest that farmers' innovative behavior and supply chain learning mediate the relationship between external pressure and value addition. The findings revealed complete mediation by farmers' innovative behavior ($\beta = 0.181, t = 5.113, p < 0.001$) and partial mediation by supply chain learning ($\beta = 0.127, t = 2.047, p < 0.001$), underscoring their critical role in supporting value addition under diverse external pressures. Hypothesis 8 suggests that higher education levels amplify the positive impact of external pressure on farmers' innovation behavior, whereas lower education levels may weaken or nullify this relationship. Education significantly shapes how external pressure influences farmers' innovative behavior, leading to variations in the adoption of innovative practices across different educational levels.

As farmer education is a categorical variable, multi-group analysis (MGA) in partial least squares path modeling (PLSPM) is recommended for assessing moderation across multiple relationships. The MGA provides a comprehensive view of the moderator's influence on all modeled relationships, ensuring a thorough analysis of its impact, especially when the moderator is categorical (Memon *et al.*, 2019). Henseler *et al.* (2015) introduced the Measurement Invariance of Composite Models (MICOM) procedure to ensure outcome validity, aligning with the latent variable scores in composite modeling in PLSPM. MGA (Multi-group Analysis) is valuable for assessing differences across groups in PLSPM (Cheah *et al.*, 2023). Neglecting heterogeneity can compromise the validity of the PLSPM and result in incorrect conclusions. Including MGA in PLSPM enhances researchers' ability to identify meaningful differences in multiple relationships across specific groups, as emphasized by Picón-Berjoto *et al.* (2016) and Schlögel and Sarstedt (2016).

The MICOM procedure involves three steps: (1) assessing configural invariance (2) evaluating compositional invariance, and (3) verifying the equality of a composite's mean value and variance across groups. If both configural and compositional invariances are confirmed, partial measurement invariance is established. Researchers can then compare path coefficients using Multi-group Analysis (MGA). Full measurement invariance allows data pooling, increases statistical power, and makes MGA unnecessary. If configural and compositional invariance are established and either composite variance or mean equality is achieved in Step III, partial measurement invariance is claimed, and MGA can proceed. The output of the bootstrapping MGA in SmartPLS 4.1.0.0 is presented in Table 7.

Table 7 outlines the multigroup bootstrapping results by comparing the path coefficients and p -values across the education levels of both groups. In the educated group, Compliance with Food Safety Standards (CWFSS) moderately affected value addition (0.234), whereas External Pressure significantly affected farmers' innovation behavior (0.684) and Supply Chain Learning (0.800). Farmers' innovation behavior (0.412) and Supply Chain Learning (0.235) significantly contribute to value addition. However, the interaction term (CWFSS $\times$ farmers' innovative behavior) was insignificant (0.056). Similar patterns were observed in the non-education group, with variable coefficients. All paths had p -values below

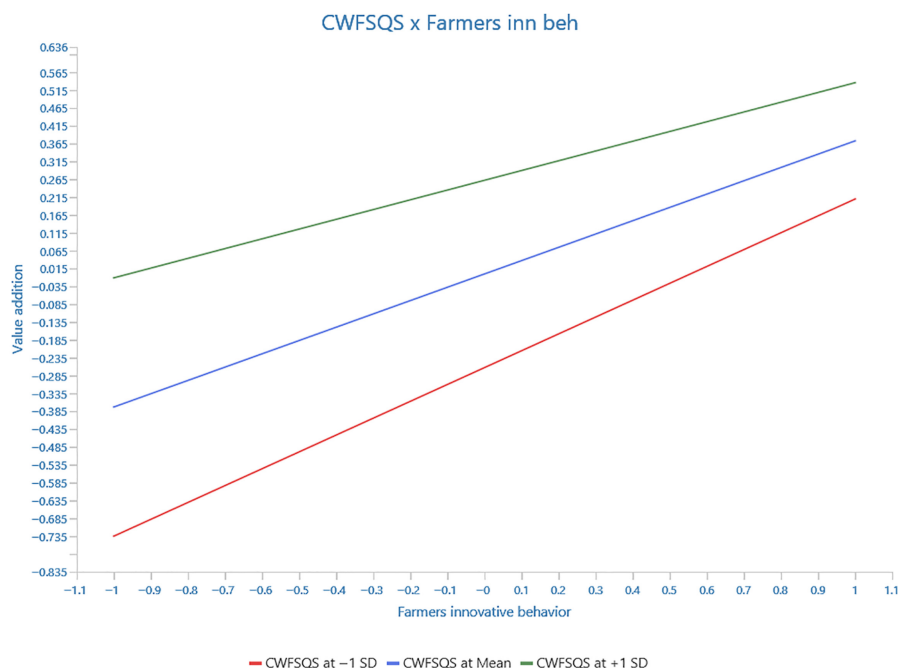
Table 7.
Path coefficients and
bootstrapping results

Path coefficient – bootstrapping results	Orig (education)	Orig (no education)	p-value (education)	p-value (no education)
CWFSS → Value addition	0.234	0.242	0.056	0.071
Ext Pressure → Farmers Innovative behavior	0.684	0.68	0.000	0.000
Ext Pressure → Supply Chain Learning	0.800	0.878	0.000	0.000
Farmers Innovative behavior → Value addition	0.412	0.392	0.000	0.000
Supply Chain Learning → Value addition	0.235	0.167	0.007	0.043
CWFSS × Farmers Innovative behavior → Value addition	0.056	0.122	0.155	0.101

Source(s): Authors' own work

0.05, indicating statistical significance in both educational groups, except for CWFSS → Value addition in the educated group and Supply Chain Learning → Value addition in the non-educated group.

The results of [Hypothesis 9](#) indicate that adherence to food safety standards significantly moderates the link between farmers' innovative behavior and value addition ($\beta = -0.103$, $t = 3.187$, $p < 0.001$), as illustrated in [Figure 4](#) through simple slope analysis. The link between

**Figure 4.**
Interaction of
compliance with food
safety standards,
farmer's innovative
behavior, and value
addition**Source(s):** Authors' own work

farmers' innovative behavior (FIN behavior), value addition (VA), and compliance with food safety regulations (CWFSS) has a negative beta value, implying that CWFSS moderates FIN behavior and value addition. A negative beta value may result from market forces that place low value on innovation and compliance, stringent compliance mandates that take resources away from innovation, and regulatory priorities that discourage risk-taking, which is essential to innovation.

Overall, the findings underscore the critical importance of food safety standards for enhancing the productivity and competitiveness of smallholder farmers in developing countries. By promoting supply chain learning and compliance with these standards, policymakers can help unlock the full potential of farmers' innovative behavior and increase value addition, thereby contributing to sustainable economic growth and poverty reduction in these countries.

5. Conclusion, practical implications, limitations, and future research directions

The empirical investigation of the intricate interactions between supply chain learning, value addition, farmers' innovative behavior, and external pressure in the context of smallholder farming closes a significant gap in the literature. External pressure was found to have a considerable influence on smallholder farmers' adoption of proactive food safety practices and the enhancement of value addition, even though its direct impact was previously considered minor.

The results demonstrate the critical role of government actions in guaranteeing food safety and encouraging value addition. Moreover, the findings highlight how farmers' innovative behavior and supply chain learning mediate the support of value addition in the face of various external challenges. Furthermore, this study highlights how compliance with food safety standards moderates the relationship between farmers' innovative behavior and value addition, indicating the need for nuanced governmental interventions to benefit smallholder farmers.

Government intervention is crucial to ensure food safety and promote value addition in the dairy industry. Policymakers should focus on providing technical support and implementing food safety regulations. Farmers can benefit from adopting innovative techniques to meet the changing external pressures, and industry players should adjust their strategies to meet stakeholder demands. Emphasizing adherence to food safety regulations is essential for realizing the benefits of novel approaches in the dairy industries of developing nations. This study provides valuable insights for policymakers on poverty reduction and sustainable economic growth.

However, limitations include a potential lack of generalizability due to focusing solely on Pakistan's dairy industry, varying effects based on agricultural contexts, and a focus on internal phenomena rather than broader external influences such as economic policy. Considering the differences in regional compliance and involving diverse stakeholders can enhance the applicability of this study to tailored interventions that support sustainable smallholder dairy farming in low-income countries. Future research could examine the entire upstream-downstream supply chain and supply chain partners, which could be beneficial in terms of helping policymaking and practically helping supply chain partners across the whole supply chain.

References

- Ab Talib, M.S., Abdul Hamid, A.B. and Chin, T.A. (2016), "Can halal certification influence logistics performance?", *Journal of Islamic Marketing*, Vol. 7 No. 4, pp. 461-475, doi: [10.1108/JIMA-02-2015-0015](https://doi.org/10.1108/JIMA-02-2015-0015).

- Abdul-Rahaman, A. and Abdulai, A. (2020), "Farmer groups, collective marketing and smallholder farm performance in rural Ghana", *Journal of Agribusiness in Developing and Emerging Economies*, Vol. 10 No. 5, pp. 511-527, doi: [10.1108/JADEE-07-2019-0095](https://doi.org/10.1108/JADEE-07-2019-0095).
- Adetoyinbo, A. and Mithöfer, D. (2023), "Inter-firm relations and resource-based performance: a contingent relational view of small-scale farmers in Zambia", *Journal of Agribusiness in Developing and Emerging Economies*, Vol. ahead-of-print No. ahead-of-print, doi: [10.1108/JADEE-06-2023-0134](https://doi.org/10.1108/JADEE-06-2023-0134).
- Agyabeng-Mensah, Y., Afum, E., Acquah, I.S.K. and Baah, C. (2023), "How does supply chain knowledge enhance green innovation? The mediation mechanisms of corporate reputation and non-supply chain learning", *Journal of Business and Industrial Marketing*, Vol. 38 No. 4, pp. 852-868, doi: [10.1108/JBIM-04-2021-0192](https://doi.org/10.1108/JBIM-04-2021-0192).
- Ahmed, W., Najmi, A., Faizan, H.M. and Ahmed, S. (2018), "Consumer behaviour towards willingness to pay for Halal products: an assessment of demand for Halal certification in a Muslim country", *British Food Journal*, Vol. 121 No. 2, pp. 492-504, doi: [10.1108/BFJ-02-2018-0085](https://doi.org/10.1108/BFJ-02-2018-0085).
- Ahmed, W., Ashraf, M.S., Khan, S.A., Kusi-Sarpong, S., Arhin, F.K., Kusi-Sarpong, H. and Najmi, A. (2020a), "Analyzing the impact of environmental collaboration among supply chain stakeholders on a firm's sustainable performance", *Operations Management Research*, Vol. 13 Nos 1-2, pp. 4-21, doi: [10.1007/s12063-020-00152-1](https://doi.org/10.1007/s12063-020-00152-1).
- Ahmed, W., Najmi, A. and Ikram, M. (2020b), "Steering firm performance through innovative capabilities: a contingency approach to innovation management", *Technology in Society*, Vol. 63, 101385, doi: [10.1016/j.techsoc.2020.101385](https://doi.org/10.1016/j.techsoc.2020.101385).
- Ahmed, W., Najmi, A. and Khan, F. (2020c), "Examining the impact of institutional pressures and green supply chain management practices on firm performance", *Management of Environmental Quality: An International Journal*, Vol. 31 No. 5, pp. 1261-1283, doi: [10.1108/MEQ-06-2019-0115](https://doi.org/10.1108/MEQ-06-2019-0115).
- Barney, J. (1991), "Firm resources and sustained competitive advantage", *Journal of Management*, Vol. 17 No. 1, pp. 99-120, doi: [10.1016/S0742-3322\(00\)17018-4](https://doi.org/10.1016/S0742-3322(00)17018-4).
- Barzola Iza, C.L., Dentoni, D. and Omta, O.S. (2020), "The influence of multi-stakeholder platforms on farmers' innovation and rural development in emerging economies: a systematic literature review", *Journal of Agribusiness in Developing and Emerging Economies*, Vol. 10 No. 1, pp. 13-39, doi: [10.1108/JADEE-12-2018-0182](https://doi.org/10.1108/JADEE-12-2018-0182).
- Becker, J.-M., Cheah, J.-H., Gholamzade, R., Ringle, C.M. and Sarstedt, M. (2023), "PLS-SEM's most wanted guidance", *International Journal of Contemporary Hospitality Management*, Vol. 35 No. 1, pp. 321-346, doi: [10.1108/IJCHM-04-2022-0474](https://doi.org/10.1108/IJCHM-04-2022-0474).
- Cao, H. and Chen, Z. (2017), "The driving effect of internal and external environment on firms' green innovation strategies-the moderating role of executives' environmental awareness", *Nankai Business Review International*, Vol. 20, pp. 95-103.
- Chaurasiya, K., Prashant Maratha, P.M. and Badodiya, S. (2017), "Factors affecting entrepreneurial behaviour of dairy farmers", *Agriculture Update*, Vol. 12 No. 1, pp. 23-30, doi: [10.15740/HAS/AU/12.1/23-30](https://doi.org/10.15740/HAS/AU/12.1/23-30).
- Cheah, J.-H., Amaro, S. and Roldán, J.L. (2023), "Multigroup analysis of more than two groups in PLS-SEM: a review, illustration, and recommendations", *Journal of Business Research*, Vol. 156, 113539, doi: [10.1016/j.jbusres.2022.113539](https://doi.org/10.1016/j.jbusres.2022.113539).
- Chen, E., Flint, S., Perry, P., Perry, M. and Lau, R. (2015), "Implementation of non-regulatory food safety management schemes in New Zealand: a survey of the food and beverage industry", *Food Control*, Vol. 47, pp. 569-576, doi: [10.1016/j.foodcont.2014.08.009](https://doi.org/10.1016/j.foodcont.2014.08.009).
- Devaux, A., Torero, M., Donovan, J. and Horton, D. (2018), "Agricultural innovation and inclusive value-chain development: a review", *Journal of Agribusiness in Developing and Emerging Economies*, Vol. 8 No. 1, pp. 99-123, doi: [10.1108/JADEE-06-2017-0065](https://doi.org/10.1108/JADEE-06-2017-0065).

- DiMaggio, P.J. and Powell, W.W. (1983), "The iron cage revisited: institutional isomorphism and collective rationality in organizational fields", *American Sociological Review*, Vol. 48 No. 2, pp. 147-160, doi: [10.1016/S0742-3322\(00\)17011-1](https://doi.org/10.1016/S0742-3322(00)17011-1).
- Doyle, E., McGovern, D. and McCarthy, S. (2014), "Compliance-innovation: integrating quality and compliance knowledge and practice", *Total Quality Management and Business Excellence*, Vol. 25 Nos 9-10, pp. 1156-1170, doi: [10.1080/14783363.2014.914641](https://doi.org/10.1080/14783363.2014.914641).
- Dunne, T.C., Aaron, J.R., McDowell, W.C., Urban, D.J. and Geho, P.R. (2016), "The impact of leadership on small business innovativeness", *Journal of Business Research*, Vol. 69 No. 11, pp. 4876-4881, doi: [10.1108/JBIM-12-2019-0534](https://doi.org/10.1108/JBIM-12-2019-0534).
- Erena, O.T., Kalko, M.M. and Debele, S.A. (2022), "Organizational factors, knowledge management and innovation: empirical evidence from medium-and large-scale manufacturing firms in Ethiopia", *Journal of Knowledge Management*, Vol. 27 No. 4, pp. 1165-1207, doi: [10.1108/JKM-11-2021-0861](https://doi.org/10.1108/JKM-11-2021-0861).
- Escanciano, C. and Santos-Vijande, M.L. (2014), "Reasons and constraints to implementing an ISO 22000 food safety management system: evidence from Spain", *Food Control*, Vol. 40, pp. 50-57, doi: [10.1016/j.foodcont.2013.11.032](https://doi.org/10.1016/j.foodcont.2013.11.032).
- FAO (2024), "Dairy market review: overview of global market developments in 2023", available at: <https://openknowledge.fao.org/items/17708737-78e8-44f3-9ab7-74e176b40f83>
- Fernando, Y., Ng, H.H. and Walters, T. (2015), "Regulatory incentives as a moderator of determinants for the adoption of Malaysian food safety system", *British Food Journal*, Vol. 117 No. 4, pp. 1336-1353, doi: [10.1108/BFJ-03-2014-0129](https://doi.org/10.1108/BFJ-03-2014-0129).
- Flores-Flores, M.E., Lizarraga, E., de Cerain, A.L. and González-Peñas, E. (2015), "Presence of mycotoxins in animal milk: a review", *Food Control*, Vol. 53, pp. 163-176, doi: [10.1016/j.foodcont.2015.01.020](https://doi.org/10.1016/j.foodcont.2015.01.020).
- Fornell, C. and Larcker, D.F. (1981), "Evaluating structural equation models with unobservable variables and measurement error", *Journal of Marketing Research*, Vol. 18 No. 1, pp. 39-50, doi: [10.1177/002224378101800104](https://doi.org/10.1177/002224378101800104).
- Gallo, M., Ferrara, L., Calogero, A., Montesano, D. and Naviglio, D. (2020), "Relationships between food and diseases: what to know to ensure food safety", *Food Research International*, Vol. 137, 109414, doi: [10.1016/j.foodres.2020.109414](https://doi.org/10.1016/j.foodres.2020.109414).
- Hair Jr, J.F., Hult, G.T.M., Ringle, C.M., Sarstedt, M., Danks, N.P. and Ray, S. (2021), *Partial Least Squares Structural Equation Modeling (PLS-SEM) Using R: A Workbook*, Springer Nature, Cham.
- Haq, M.Z.U. (2021), "Supply chain learning and organizational performance: evidence from Chinese manufacturing firms", *Journal of Knowledge Management*, Vol. 25 No. 4, pp. 943-972, doi: [10.1108/JKM-05-2020-0335](https://doi.org/10.1108/JKM-05-2020-0335).
- Haq, M.Z.U., Gu, M. and Huo, B. (2021), "Enhancing supply chain learning and innovation performance through human resource management", *Journal of Business and Industrial Marketing*, Vol. 36 No. 3, pp. 552-568, doi: [10.1108/JBIM-12-2019-0534](https://doi.org/10.1108/JBIM-12-2019-0534).
- Henseler, J., Ringle, C. and Sarstedt, M. (2015), "A new criterion for assessing discriminant validity in variance-based structural equation modeling", *Journal of the Academy of Marketing Science*, Vol. 43 No. 1, pp. 115-135, doi: [10.1007/s11747-014-0403-8](https://doi.org/10.1007/s11747-014-0403-8).
- Huang, L., Zhou, Y., Han, Y., Hammitt, J.K., Bi, J. and Liu, Y. (2013), "Effect of the Fukushima nuclear accident on the risk perception of residents near a nuclear power plant in China", *Proceedings of the National Academy of Sciences*, Vol. 110 No. 49, pp. 19742-19747, doi: [10.1073/pnas.1313825110](https://doi.org/10.1073/pnas.1313825110).
- Huo, B., Haq, M.Z.U. and Gu, M. (2019), "The impact of IT application on supply chain learning and service performance", *Industrial Management and Data Systems*, Vol. 120 No. 1, pp. 1-20, doi: [10.1108/IMDS-10-2018-0476](https://doi.org/10.1108/IMDS-10-2018-0476).

- Jaeger, S.R., Antúnez, L. and Ares, G. (2023), "An exploration of what freshness in fruit means to consumers", *Food Research International*, Vol. 165, 112491, doi: [10.1016/j.foodres.2023.112491](https://doi.org/10.1016/j.foodres.2023.112491).
- Jatib, I. (2003), "Food safety and quality assurance key drivers of competitiveness", *International Food and Agribusiness Management Review*, Vol. 6 No. 1, pp. 1-19, doi: [10.22004/ag.econ.34302](https://doi.org/10.22004/ag.econ.34302).
- Khan, M., Najmi, A., Ahmed, W. and Aman, A. (2019), "The role of consumer willingness to pay for halal certification in Pakistan", *Journal of Islamic Marketing*, Vol. 10 No. 4, pp. 1230-1244, doi: [10.1108/JIMA-09-2018-0155](https://doi.org/10.1108/JIMA-09-2018-0155).
- Kock, N. (2015), "Common method bias in PLS-SEM: a full collinearity assessment approach", *International Journal of e-Collaboration (ijec)*, Vol. 11 No. 4, pp. 1-10, doi: [10.4018/ijec.2015100101](https://doi.org/10.4018/ijec.2015100101).
- Korir, L., Manning, L., Moore, H.L., Lindahl, J.F., Gemechu, G., Mihret, A., Berg, S., Wood, J.L.N. and Nyokabi, N.S. (2023), "Adoption of dairy technologies in smallholder dairy farms in Ethiopia", *Frontiers in Sustainable Food Systems*, Vol. 7, 1070349, doi: [10.3389/fsufs.2023.1070349](https://doi.org/10.3389/fsufs.2023.1070349).
- Kurihara, S., Ishida, T., Maruyama, A., Luloff, A. and Kanayama, T. (2014), "Role of risk-related latent factors in the adoption of new production technology the case of Japanese greenhouse vegetable farmers", *International Journal of Agricultural Science and Technology*, Vol. 2 No. 2, pp. 53-60, doi: [10.14355/ijast.2014.0302.01](https://doi.org/10.14355/ijast.2014.0302.01).
- Laguna, M., Mielniczuk, E. and Gorgievski, M.J. (2021), "Business owner-employees contagion of work-related affect and employees' innovative behavior in small firms", *Applied Psychology*, Vol. 70 No. 4, pp. 1543-1571, doi: [10.1111/apps.12288](https://doi.org/10.1111/apps.12288).
- Lewison, E. (2022), "Plant, sleep, pick': ambivalent smallholder market engagement and inclusive value chains in northwest Nepal", *Development and Change*, Vol. 53 No. 4, pp. 860-887, doi: [10.1111/dech.12722](https://doi.org/10.1111/dech.12722).
- Liu, Z., Zhang, Y. and Li, H. (2021), "Digital inclusive finance, multidimensional education, and farmers' entrepreneurial behavior", *Mathematical Problems in Engineering*, Vol. 2021, pp. 1-13, 2021 doi: [10.1155/2021/6541437](https://doi.org/10.1155/2021/6541437).
- Lund-Thomsen, P., Riisgaard, L., Singh, S., Ghorri, S. and Coe, N.M. (2021), "Global value chains and intermediaries in Multi-stakeholder initiatives in Pakistan and India", *Development and Change*, Vol. 52 No. 3, pp. 504-532, doi: [10.1111/dech.12647](https://doi.org/10.1111/dech.12647).
- Mandala, G.N., Verma, M., Verma, A., Sirisetti, S. and Gandreti, V.R.R. (2021), "The influence of agricultural farmers' entrepreneurial behavior on the business performance of dairy farmers in Andhra Pradesh", *Universal Journal of Agricultural Research*, Vol. 9 No. 2, pp. 57-61, doi: [10.13189/ujar.2021.090204](https://doi.org/10.13189/ujar.2021.090204).
- Manjara, A.M. (2021), "Educational level and adoption of process innovation in the dairy sector of Zimbabwe", *International Journal of Engineering Applied Sciences and Technology*, Vol. 5 No. 9, pp. 116-118, doi: [10.33564/IJEAST.2021.V05I09.016](https://doi.org/10.33564/IJEAST.2021.V05I09.016).
- Matzembacher, D.E. and Meira, F.B. (2019), "Sustainability as business strategy in community supported agriculture: social, environmental and economic benefits for producers and consumers", *British Food Journal*, Vol. 121 No. 2, pp. 616-632, doi: [10.1108/BFJ-03-2018-0207](https://doi.org/10.1108/BFJ-03-2018-0207).
- Mellon Bedi, S., Kornher, L., von Braun, J. and Kotu, B.H. (2022), "Stimulating innovations for sustainable agricultural practices among smallholder farmers: persistence of intervention matters", *The Journal of Development Studies*, Vol. 58 No. 9, pp. 1651-1667, doi: [10.1080/00220388.2022.2043283](https://doi.org/10.1080/00220388.2022.2043283).
- Memon, M.A., Cheah, J.-H., Ramayah, T., Ting, H., Chuah, F. and Cham, T.H. (2019), "Moderation analysis: issues and guidelines", *Journal of Applied Structural Equation Modeling*, Vol. 3 No. 1, pp. 1-11, doi: [10.47263/jasem.3\(1\)01](https://doi.org/10.47263/jasem.3(1)01).
- Mottet, A., Bicksler, A., Lucantoni, D., De Rosa, F., Scherf, B., Scopel, E., López-Ridaura, S., Gemmil-Herren, B., Bezner Kerr, R., Petersen, P., Chotte, J.L., Loconto, A., Titttonell, P. and Sourisseau, J.-M. (2020), "Assessing transitions to sustainable agricultural and food systems: a tool for agroecology performance evaluation (TAPE)", *Frontiers in Sustainable Food Systems*, Vol. 4, 579154, doi: [10.3389/fsufs.2020.579154](https://doi.org/10.3389/fsufs.2020.579154).

- Mugonya, J., Kalule, S. and Ndyomugenyi, E. (2021), "Effect of market information quality, sharing and utilisation on the innovation behaviour of smallholder pig producers", *Cogent Food and Agriculture*, Vol. 7 No. 1, 1948726, doi: [10.1080/23311932.2021.1948726](https://doi.org/10.1080/23311932.2021.1948726).
- Mukhtar, S., Gwazawa, U. and Jega, A. (2018), "Entrepreneurship development for diversification of Nigerian economy", *Journal of Economics, Management and Trade*, Vol. 21 No. 6, pp. 1-11, doi: [10.9734/JEMT/2018/41679](https://doi.org/10.9734/JEMT/2018/41679).
- Najmi, A., Ahmed, W. and Jahangir, S. (2023), "Firm's readiness for halal food standard adoption: assessing the importance of traceability system", *Journal of Islamic Accounting and Business Research*, Vol. 14 No. 8, pp. 1451-1473, doi: [10.1108/JIABR-09-2020-0293](https://doi.org/10.1108/JIABR-09-2020-0293).
- Negi, S. and Anand, N. (2019), "Wholesalers perspectives on mango supply chain efficiency in India", *Journal of Agribusiness in Developing and Emerging Economies*, Vol. 9 No. 2, pp. 175-200, doi: [10.1108/JADEE-02-2018-0032](https://doi.org/10.1108/JADEE-02-2018-0032).
- Negi, S. and Trivedi, S. (2021), "Factors impacting the quality of fresh produce in transportation and their mitigation strategies: empirical evidence from a developing economy", *Journal of Agribusiness in Developing and Emerging Economies*, Vol. 11 No. 2, pp. 121-139, doi: [10.1108/JADEE-07-2020-0154](https://doi.org/10.1108/JADEE-07-2020-0154).
- Nguyen-Anh, T. (2021), "Market-oriented extension and farming efficiency in small-scale maize farmers: evidence from Northern Vietnam", *Journal of Agribusiness in Developing and Emerging Economies*, Vol. 11 No. 2, pp. 194-218, doi: [10.1108/JADEE-05-2020-0101](https://doi.org/10.1108/JADEE-05-2020-0101).
- Nk, T., Uwimana, P., Majiwa, E. and Niyitanga, F. (2023), "Impact of contract farming on the income of smallholder dairy farmers from nyagatare district in the eastern province of Rwanda", *African Journal of Food, Agriculture, Nutrition and Development*, Vol. 23 No. 5, pp. 23465-23488, doi: [10.18697/ajfand.120.22745](https://doi.org/10.18697/ajfand.120.22745).
- Nurlaela, S., Raya, A.B. and Hariadi, S.S. (2022), "Information technology utilization of young educated farmers in agricultural entrepreneurship", *Agro Ekonomi*, Vol. 33 No. 1, pp. 11-21, doi: [10.22146/ae.64524](https://doi.org/10.22146/ae.64524).
- Nyokabi, S., Luning, P.A., de Boer, I.J., Korir, L., Muunda, E., Bebe, B.O., Lindahl, J., Bett, B. and Oosting, S.J. (2021), "Milk quality and hygiene: knowledge, attitudes and practices of smallholder dairy farmers in central Kenya", *Food Control*, Vol. 130, pp. 1-10, 108303, doi: [10.1016/j.foodcont.2021.108303](https://doi.org/10.1016/j.foodcont.2021.108303).
- Ojha, D., Acharya, C. and Cooper, D. (2018), "Transformational leadership and supply chain ambidexterity: mediating role of supply chain organizational learning and moderating role of uncertainty", *International Journal of Production Economics*, Vol. 197, pp. 215-231, doi: [10.1016/j.ijpe.2018.01.001](https://doi.org/10.1016/j.ijpe.2018.01.001).
- Owot, G.M., Okello, D.M., Olido, K. and Odongo, W. (2023), "Trust, but what trust? Investigating the influence of trust dimensions on supply chain performance in smallholder agribusinesses in Uganda", *Journal of Agribusiness in Developing and Emerging Economies*, Vol. ahead-of-print No. ahead-of-print, doi: [10.1108/JADEE-09-2022-0196](https://doi.org/10.1108/JADEE-09-2022-0196).
- Park, H.Y., Misra, K., Reddy, S. and Jaber, K. (2019), "Family firms' innovation drivers and performance: a dynamic capabilities approach", *Journal of Family Business Management*, Vol. 9 No. 1, pp. 4-23, doi: [10.1108/JFBM-11-2017-0039](https://doi.org/10.1108/JFBM-11-2017-0039).
- Penwell, M.O., Sam, N.T., Tshepo, R. and Matshonisa, S.S. (2023), "Potential benefits of crowd-collaborative farming in sustaining small scale dairy farming business performance", *International Journal of Entrepreneurship and Sustainability Studies*, Vol. 3 No. 2, pp. 110-124, doi: [10.31098/ijeass.v3i2.1972](https://doi.org/10.31098/ijeass.v3i2.1972).
- Picón-Berjoyo, A., Ruiz-Moreno, C. and Castro, I. (2016), "A mediating and multigroup analysis of customer loyalty", *European Management Journal*, Vol. 34 No. 6, pp. 701-713, doi: [10.1016/j.emj.2016.07.006](https://doi.org/10.1016/j.emj.2016.07.006).
- Ramayah, T., Cheah, J., Chuah, F., Ting, H. and Memon, M.A. (2018), *Partial Least Squares Structural Equation Modeling (PLS-SEM) Using smartPLS 3.0: An Updated Guide And Practical Guide To Statistical Analysis*, 2nd ed., Kuala Lumpur, Pearson.

- Roussy, C., Ridier, A. and Chaib, K. (2017), "Farmers' innovation adoption behaviour: role of perceptions and preferences", *International Journal of Agricultural Resources, Governance and Ecology*, Vol. 13 No. 2, pp. 138-161, doi: [10.1504/IJARGE.2017.086439](https://doi.org/10.1504/IJARGE.2017.086439).
- Sarstedt, M., Hair, J.F. and Ringle, C.M. (2023), "PLS-SEM: indeed a silver bullet'-retrospective observations and recent advances", *Journal of Marketing Theory and Practice*, Vol. 31 No. 3, pp. 261-275, doi: [10.1080/10696679.2022.2056488](https://doi.org/10.1080/10696679.2022.2056488).
- Sarwar, M., Khan, M., Mahr-un-Nisa, M.-u.-N. and Iqbal, Z. (2002), "Dairy industry in Pakistan: a scenario", *International Journal of Agriculture and Biology*, Vol. 4 No. 3, pp. 420-428.
- Schiller, K.J., Klerkx, L., Poortvliet, P.M. and Godek, W. (2020), "Exploring barriers to the agroecological transition in Nicaragua: a technological innovation systems approach", *Agroecology and Sustainable Food Systems*, Vol. 44 No. 1, pp. 88-132, doi: [10.1080/21683565.2019.1602097](https://doi.org/10.1080/21683565.2019.1602097).
- Schlägel, C. and Sarstedt, M. (2016), "Assessing the measurement invariance of the four-dimensional cultural intelligence scale across countries: a composite model approach", *European Management Journal*, Vol. 34 No. 6, pp. 633-649, doi: [10.1016/j.emj.2016.06.002](https://doi.org/10.1016/j.emj.2016.06.002).
- Shea, C.M., Jacobs, S.R., Esserman, D.A., Bruce, K. and Weiner, B.J. (2014), "Organizational readiness for implementing change: a psychometric assessment of a new measure", *Implementation Science*, Vol. 9 No. 1, pp. 1-15, doi: [10.1186/1748-5908-9-7](https://doi.org/10.1186/1748-5908-9-7).
- Shen, B., Choi, T.-M. and Minner, S. (2019), "A review on supply chain contracting with information considerations: information updating and information asymmetry", *International Journal of Production Research*, Vol. 57 Nos 15-16, pp. 4898-4936, doi: [10.1080/00207543.2018.1467062](https://doi.org/10.1080/00207543.2018.1467062).
- Snorek, J., Cummings, W., Hryniewicz, E., Stevens, K. and Iannuzzi, R. (2023), "Diversification strategies for the resilience of small New England dairies", *Journal of Agriculture, Food Systems, and Community Development*, Vol. 12 No. 3, pp. 1-21, doi: [10.5304/jafscd.2023.123.004](https://doi.org/10.5304/jafscd.2023.123.004).
- Tahir, M.N., Riaz, R., Bilal, M. and Nouman, H.M. (2019), "Current standing and future challenges of dairying in Pakistan: a status update", in Javed, K. (Ed.), *Milk Production, Processing and Marketing*, Intech Open, London, pp. 1-24.
- Takács-György, K. (2016), *Innovation for food safety*.
- Tan, Y., Sarkar, A., Rahman, A., Qian, L., Hussain Memon, W. and Magzhan, Z. (2021), "Does external shock influence farmer's adoption of modern irrigation technology?—a case of Gansu Province, China", *Land*, Vol. 10 No. 8, p. 882, doi: [10.3390/land10080882](https://doi.org/10.3390/land10080882).
- Tavana, M., Shabani, A. and Singh, R. (2019), "The impact of interwoven integration practices on supply chain value addition and firm performance", *Journal of Industrial Engineering International*, Vol. 15 No. S1, pp. 39-51, doi: [10.1007/s40092-019-0316-8](https://doi.org/10.1007/s40092-019-0316-8).
- Tebeka, Y., Katungi, E., Rubyogo, J.-C., Sserunkuuma, D. and Kidane, T. (2017), "Economic performance of community based bean seed production and marketing in the central rift valley of Ethiopia", *African Crop Science Journal*, Vol. 25 No. 2, pp. 189-205, doi: [10.4314/acsj.v25i2.5](https://doi.org/10.4314/acsj.v25i2.5).
- Thöni, A. and Tjoa, A.M. (2017), "Information technology for sustainable supply chain management: a literature survey", *Enterprise Information Systems*, Vol. 11 No. 6, pp. 828-858, doi: [10.1080/17517575.2015.1091950](https://doi.org/10.1080/17517575.2015.1091950).
- Walder, P., Sinabell, F., Unterlass, F., Niedermayr, A., Fulgeanu, D., Kapfer, M., Melcher, M. and Kantelhardt, J. (2019), "Exploring the relationship between farmers' innovativeness and their values and aims", *Sustainability*, Vol. 11 No. 20, p. 5571, doi: [10.3390/su11205571](https://doi.org/10.3390/su11205571).
- Wangu, J., Mangnus, E. and van Westen, A. (2021), "Recognizing determinants to smallholders' market orientation and marketing arrangements: building on a case of dairy farming in rural Kenya", *Land*, Vol. 10 No. 6, p. 572, doi: [10.3390/land10060572](https://doi.org/10.3390/land10060572).
- Wegerif, M.C. and Martucci, R. (2019), "Milk and the city: raw milk challenging the value claims of value chains", *Agroecology and Sustainable Food Systems*, Vol. 43 No. 10, pp. 1077-1105, doi: [10.1080/21683565.2018.1530716](https://doi.org/10.1080/21683565.2018.1530716).

- Wongnaa, C.A., Adams, F., Bannor, R.K., Awunyo-Vitor, D., Mahama, I., Osei, B.A., Owusu-Ansah, Y. and Ackon, A. (2019), "Job creation and improved consumer health through commercialisation of tiger nut yoghurt: a willingness to pay analysis", *Journal of Global Entrepreneurship Research*, Vol. 9, pp. 1-22, doi: [10.1080/17517575.2015.1091950](https://doi.org/10.1080/17517575.2015.1091950).
- Xhoxhi, O., Dentoni, D., Imami, D., Skreli, E. and Sokoli, O. (2019), "Does farmers' trust towards buyers drive their entrepreneurial orientation? Evidence from a transition economy", *World Development*, Vol. 12 No. 1, pp. 142-159, (Forthcoming) doi: [10.1108/JADEE-03-2020-0045](https://doi.org/10.1108/JADEE-03-2020-0045).
- Yang, Y., Jia, F. and Xu, Z. (2019), "Towards an integrated conceptual model of supply chain learning: an extended resource-based view", *Supply Chain Management: An International Journal*, Vol. 24 No. 2, pp. 189-214, doi: [10.1108/SCM-11-2017-0359](https://doi.org/10.1108/SCM-11-2017-0359).
- Yin, Y., Zhang, Y., Li, F., Jiao, J., Lebailly, P., Zhang, Y. and Yin, C. (2022), "Driving mechanism for farmers' participation in improving farmland ecosystem: evidence from China", *Journal of Cleaner Production*, Vol. 380, 134895, doi: [10.1016/j.jclepro.2022.134895](https://doi.org/10.1016/j.jclepro.2022.134895).
- Zhang, H.-Y. and Lv, S. (2015), "Intellectual capital and technological innovation: the mediating role of supply chain learning", *International Journal of Innovation Science*, Vol. 7 No. 3, pp. 199-210, doi: [10.1260/1757-2223.7.3.199](https://doi.org/10.1260/1757-2223.7.3.199).

Corresponding author

Muhammad Moazzam can be contacted at: moazzam.m.sial@gmail.com